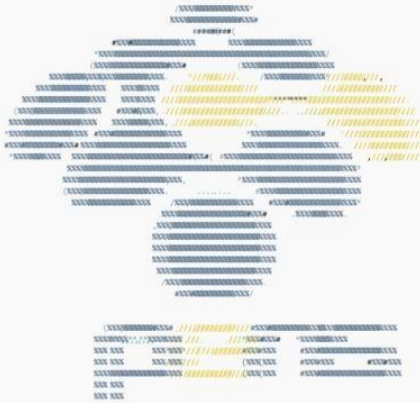




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P5- Pemrograman Statement Perulangan Looping

- - DISUSUN OLEH - -

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P5- Pemrograman Statement Perulangan Looping

Tugas 5.1 Matriks Berulang

Kode Program

```
1  #include <stdio.h>
2
3  int main() {
4      int baris, kolom;
5
6      printf("masukkan baris =");
7      scanf("%d",&baris);
8      printf("masukkan kolom = ");
9      scanf("%d",&kolom);
10     printf("\n");
11
12     for (int i = 1; i < baris; i++) {
13         for (int j = 1; j <= kolom; j++) {
14             printf("%d\t", j);
15         }
16         printf("\n");
17     }
18
19     return 0;
20 }
```

Output

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

Tugas 5.2 Pangkat Matriks

Kode Program

```
1  #include <stdio.h>
2
3  int main() {
4      int baris, kolom, hasil;
5      printf("masukkan baris = ");
6      scanf("%d",&baris);
7      printf("masukkan kolom = ");
8      scanf("%d",&kolom);
9      printf("\n");
10
11     for (int i = 1; i <= baris; i++) {
12         for (int j = 1; j <= kolom; j++) {
13             hasil = 1;
14             for (int k = 1; k <= i; k++) {
15                 hasil *= j;
16             }
17             printf("%9d", hasil);
18         }
19         printf("\n");
20     }
21     return 0;
22 }
```

Output

1	2	3	4	5
1	4	9	16	25
1	8	27	64	125
1	16	81	256	625
1	32	243	1024	3125

Tugas 5.3 Deret Bilangan

Kode Program

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      printf("Masukkan jumlah suku (n): ");
6      scanf("%d", &n);
7      printf("1) Pola 2^n: ");
8      int pola1 = 1;
9      for (int i = 1; i <= n; i++) {
10         pola1 *= 2;
11         printf("%d", pola1);
12         if (i < n) printf(", ");
13     }
14     printf("\n");
15     printf("2) Pola n^2: ");
16     for (int i = 1; i <= n; i++) {
17         printf("%d", i * i);
18         if (i < n) printf(", ");
19     }
20     printf("\n");
21     printf("3) Pola n^3: ");
22     for (int i = 1; i <= n; i++) {
23         printf("%d", i * i * i);
24         if (i < n) printf(", ");
25     }
26     printf("\n");
27     return 0;
28 }
```

Output

```
1) Pola 2^n: 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024
2) Pola n^2: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100
3) Pola n^3: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000
```

Tugas 5.4 Desimal ke Biner

Kode Program

```
1  #include <stdio.h>
2  int main ()
3  {
4      int bil, sisa, hasil;
5      printf("masukkan bilangan = ");
6      scanf("%d",&bil );
7      do {
8          sisa = bil % 2;
9          hasil = bil / 2;
10         bil = hasil ;
11         printf("hasil=%d sisa=%d\n\n", bil, sisa);
12     } while (hasil>0);
13 }
```

Output

```
masukkan bilangan = 25
hasil=12 sisa=1

hasil=6 sisa=0

hasil=3 sisa=0

hasil=1 sisa=1

hasil=0 sisa=1
```

Tugas 5.5 Bilangan Prima

Kode Program

```
1  #include <stdio.h>
2
3  int main() {
4      int max;
5      int count_prima = 0;
6      int total_sum = 0;
7
8      printf("input nilai maksimum: ");
9      scanf("%d", &max);
10     printf("Output: ");
11     for (int i = 2; i <= max; i++) {
12         int is_prime = 1;
13         for (int j = 2; j * j <= i; j++) {
14             if (i % j == 0) {
15                 is_prime = 0;
16                 break;
17             }
18         }
19         if (is_prime) {
20             if (count_prima > 0) {
21                 printf(", ");
22             }
23             printf("%d", i);
24             count_prima++;
25             total_sum += i;
26         }
27     }
28     printf("\n");
29     printf("Jumlah bilangan Prima = %d\n", count_prima);
30     printf("Jumlah seluruh bilangan Prima = %d\n", total_sum);
31     return 0;
32 }
```

Output

```
input nilai maksimum: 100
Output: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97
Jumlah bilangan Prima = 25
Jumlah seluruh bilangan Prima = 1060
```

Tugas 5.6 Iterasi Akar Kuadrat

Kode Program

```
1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      double n, akar, baru;
6      int iterasi = 0;
7      printf("Masukkan bilangan: ");
8      scanf("%lf", &n);
9      if (n < 0) {
10         printf("Bilangan tidak boleh negatif.\n");
11         return 1;
12     }
13     if (n == 0) {
14         printf("Akar = 0\n");
15         return 0;
16     }
17     akar = n / 2;
18     while (1) {
19         baru = (akar + n / akar) / 2;
20         iterasi++;
21         printf("Iterasi %d: Akar = (%.4f + %.4f/%.4f)/2 = %.4f\n",
22             iterasi, akar, n, akar, baru);
23         if (round(baru * 10000) == round(akar * 10000)) {
24             akar = baru;
25             break;
26         }
27         akar = baru;
28     }
29     double bulat = round(akar);
30     if (bulat * bulat == n) {
31         printf("\nHasil akhir: %.0f (bilangan cantik, kuadrat sempurna)\n", bulat);
32     } else {
33         printf("\nHasil akhir: %.4f\n", akar);
34     }
35     return 0;
36 }
```

Output

```
Masukkan bilangan: 46
Iterasi 1: Akar = (23.0000 + 46.0000/23.0000)/2 = 12.5000
Iterasi 2: Akar = (12.5000 + 46.0000/12.5000)/2 = 8.0900
Iterasi 3: Akar = (8.0900 + 46.0000/8.0900)/2 = 6.8880
Iterasi 4: Akar = (6.8880 + 46.0000/6.8880)/2 = 6.7831
Iterasi 5: Akar = (6.7831 + 46.0000/6.7831)/2 = 6.7823
Iterasi 6: Akar = (6.7823 + 46.0000/6.7823)/2 = 6.7823

Hasil akhir: 6.7823
```